Changmin Lee

M.S. Student, KAIST Visual Media Lab

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Google Scholar / LinkedIn / GitHub

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Korea
M.S. in Culture Technology — *Mar. 2025 - Present*
Graduate School of Culture Technology; GPA: 3.93/4.3. Advisor: Prof. Junyong Noh.

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Daegu, Korea
B.S. in Convergence Engineering — *Mar. 2018 - Feb. 2025*
School of Undergraduate Studies, College of Transdisciplinary Studies; GPA: 3.5/4.3.

University of California, Berkeley — Berkeley, CA, USA
Exchange Student; Coursework in Statistics — *Jun. 2019 - Aug. 2019*

Research Interests

- Bidirectional knowledge transfer between 2D generative models and 3D graphics.
- Generative digital avatars: creation, stylization, editing, and facial identity.

Publications

- **Editability Map-Guided Morphing for 4D Gaussian Head Avatar Stylization**
  Changmin Lee, Hyeonho Na, Kwan Yun, and Junyong Noh.
  Under Review.
- **StyleID: A Perception-Aware Dataset and Metric for Stylization-Agnostic Facial Identity Recognition**
  Kwan Yun, **Changmin Lee**, Ayeong Jeong, Youngseo Kim, Seungmi Lee, and Junyong Noh.
  ACM Transactions on Graphics (SIGGRAPH 2026), 2026.

Experience

Visual Media Lab, KAIST — Daejeon, Korea
Research Assistant; Advisor: Prof. Junyong Noh — *Apr. 2025 - Present*
Research on generative digital avatars, head avatar stylization, and facial identity.

Visual Media Lab, KAIST — Daejeon, Korea
Research Intern; Advisor: Prof. Junyong Noh — *Jun. 2024 - Aug. 2024*
Researched full-body avatar texture creation.

Klapoo
Co-Founder and CEO — *Dec. 2021 - May 2024*
Operated Daengbuhae, a multi-platform pet cosmetics curation service with 8,000+ products and 14,000+ ingredients, supporting ingredient-based safety assessments using veterinary standards.

Skills

- **Languages**: English: Professional working proficiency; Korean: Native or bilingual proficiency
- **Programming Languages**: Python, C++
- **Frameworks**: OpenGL, OpenCV, PyTorch, TensorFlow, Keras

Projects

IITP SW Starlab
Generating Hyper-Realistic to Extremely-stylized Face Avatar with Varied Speech Speed and Context-based Emotional Expression
Vice Project Manager: Oct. 2026 - Present; Project Manager: scheduled from 2027.
Research Contributions

- **Photorealistic 3D Human Facial Model Generation from a Single Image**
  SSIM ↑: 0.875 → 0.887. Mapped a single facial image into a standardized latent space to initialize a 3D face; estimated pixel-wise 3D spatial positions to enhance fine-grained facial detail.
- **Animatable Gaussian Splatting-based Photorealistic 3D Human Face Reconstruction from Video**
  SSIM ↑: 0.930 → 0.935. Initialized Gaussian Splatting by fitting and refining a template mesh on facial video; separated face and background, extracted key facial features, and reconstructed photorealistic 3D faces.

Additional Information

Military service: Auxiliary Police, Jun. 2020 - Dec. 2021; full-term discharge as Sugyeong (Sergeant equivalent). Security Unit, Daegu Metropolitan Police Agency; Crime Prevention Patrol Unit, Daegu Suseong Police Station.

References

Junyong Noh, Professor, Graduate School of Culture Technology, KAIST
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